NANYANG TECHNOLOGICAL UNIVERSITY

SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2 MH1301 Discrete Mathematics Tutorial 2

The following four questions are your warm-up exercises. We will not be discussing them during tutorial.

Ex. 6.2.2. Use the (generalized) pigeonhole principle to show that if there are 30 students in a class, then at least two have last names that begin with the same letter.

Solution: There are 26 English letters, i.e., pigeonholes, and there are 30 students, i.e., pigeons, hence there are more pigeons than pigeonholes (i.e., $30 > 26$), hence there will be least $\lceil \frac{30}{26} \rceil = 2$ students sharing the same letter.

Additional Question. 250 courses have their final exams scheduled over a period of 20 days. There are 3 possible time slots in which an exam can be held on a given day. At most two exams can take place in the same venue at the same time. Show that at least 3 venues need to be booked.

Solution: The maximum number of exams a venue can hold over the period of 20 days is

$$20 \times 3 \times 2 = 120.$$

By the generalized pigeonhole principle, the minimum number of venues needed to be booked is

$$\left\lceil \frac{250}{120} \right\rceil = 3.$$

Ex. 6.3.3. Given $S = \{a, b, c, d, e, f, g\}$, how many 5-permutations end with a? How many 5-combinations contain a?

Solution: $P(6, 4) = 360$. $C(6, 4) = 15$.

Ex. 6.3.27. How many bit strings of length 10 contain at least three 1's and at least three 0's?

Solution: The number of 1s can be 3, 4, 5, 6, or 7, hence the answer is

$$C(10, 3) + C(10, 4) + C(10, 5) + C(10, 6) + C(10, 7) = 912.$$

For the tutorial on 1 February, let's discuss the following questions (as many as time permits).

Ex. 6.2.4, 6.2.8, 6.2.32, Additional Question, 6.3.7, 6.3.8, 6.3.13, 6.3.16, 6.3.19, 6.3.25.

Ex. 6.2.4. A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.

(a) How many balls must she select to be sure of having at least three balls of the same color?
(b) How many balls must she select to be sure of having at least three blue balls?

Solution:

(a) Let n be the number of balls (i.e., pigeons) needed. There are $k = 2$ colors, i.e., pigeonholes. In order to have $\lceil n/k \rceil = \lceil n/2 \rceil = 3$, the minimum $n = 5$, since $\lceil 5/2 \rceil = 3$ and $\lceil 4/2 \rceil = 2 < 3$.
(b) She must select 13 balls. The worst case happens when she selects 10 red balls first, followed by 3 blue balls.

Ex. 6.2.8. Let $(x_i, y_i), i = 1, 2, 3, 4, 5$, be a set of five distinct points with integer coordinates in the xy plane. Show that the midpoint of the line joining at least one pair of these points has integer coordinates.

Solution: First, we note for any two integers x and y, the average, $(x + y)/2$ is again an integer if and only if both x and y are of the same parity (i.e. both odd or both even). Using this fact, it suffices to examine the x- and y-coordinates modulo 2.

Consider $(x_i \pmod 2), y_i \pmod 2))$, there will be $2 \times 2 = 4$ possible values $(0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$, i.e., 4 pigeonholes, and there are 5 points, i.e., pigeons. There are more pigeons than pigeonholes, hence at least two points will be falling in the same pigeonhole, the midpoint of these two points will be of integer coordinates.

Ex. 6.2.32. Prove that at a party where there are at least two people, there are two people who know the same number of other people there. (Assume knowledge is mutual, e.g., if A knows B, then B knows A as well.)

Solution: Consider two cases.

Case 1. There is someone, call it A, who knows everybody else in the party, i.e., everybody else knows A. Hence everybody in the party knows at least one person. Note the maximum number of people one can know is $n - 1$, this is the case for A since he/she knows all the other $n - 1$ people in the party. Hence, there are n people, pigeons, and $n - 1$ possible numbers ($[1, n - 1]$) of people each person can know, i.e., pigeonholes, hence there will be at least two people who know the same number of people.

Case 2. There is no one who knows all the other $n - 1$ people. Then the possible number of people anyone knows is in the range $[0, n - 2]$, and again there are only $n - 1$ possibilities, i.e., pigeonhole. The number of pigeon remains as n, hence conclusion holds for this case as well.

Additional Question. Assume the relation between any two people is either friend or stranger, prove that among any party of 6 people, there exist three people who are all friends to one another, or all strangers to one another.

Solution: Let's name the 6 people A, B, C, D, E, F. Consider the relation between A and the other 5 people, i.e., 5 pigeons. There are 2 different relations, i.e., pigeonholes. Hence, by the generalized pigeonhole principle, there will be at least $\lceil 5/2 \rceil = 3$ relations of the same type. Without loss of generality, assume these 3 people are B, C, D and the relation is stranger.

If B and C are strangers, then A, B, C are three people who are strangers to one another, and we are done. So it suffices to consider the case when B and C are friends. The same reasoning shows that it suffices to consider the case when B and D are friends, and C and D are friends. Therefore, B, C, and D are three friends to one another.

Consider the relations between A, B, C, (B, C) should NOT be stranger (i.e., they should be friends), since if it was then A, B, C are all strangers to each other, then we are done. Similarly, (C, D) are friends and (B, D) are friends as well. But then we end up with (B, C, D) all friends with each other.

Ex. 6.3.7. How many bit strings of length 10 contain

(a) exactly four 1's?
(b) at most four 1's?
(c) at least four 1's?
(d) an equal number of 0's and 1's?

Solution:

(a) $C(10, 4) = 210$.
(b) $C(10, 4) + C(10, 3) + C(10, 2) + C(10, 1) + C(10, 0) = 386$.
(c) $C(10, 4) + C(10, 5) + C(10, 6) + C(10, 7) + C(10, 8) + C(10, 9) + C(10, 10) = 848$, or $2^{10} - (C(10, 0) + C(10, 1) + C(10, 2) + C(10, 3)) = 848$.
(d) $C(10, 5) = 252$.

Ex. 6.3.8. A group contain n men and n women. How many ways are there to arrange these people in a row if the men and women alternate?

Solution: Among the men, there are $P(n, n)$ ways to arrange them, and similarly there are $P(n, n)$ ways to arrange the women.Either man or woman can go first, i.e., 2 choices. Thus, by product role there are $2(n!)^2$ ways.

Ex. 6.3.13. How many permutations of the letters $ABCDEFG$ contain

(a) the string BCD?
(b) the string $CFGA$?
(c) the strings BA and GF?
(d) the strings ABC and DE?
(e) the strings ABC and CDE?
(f) the strings CBA and BED?

Solution:

(a) We treat BCD as a unit, together with the other 4 characters, there are 5 units, hence $P(5, 5) = 5! = 120$ permutations.
(b) Similar to part 1, the answer is $P(4, 4) = 4! = 24$.
(c) Similar to part 1, we treat BA as one unit, and GF as another unit, and together with the remaining characters, there will be 5 units, hence $P(5, 5) = 5! = 120$ permutations.
(d) Similar to part 3, $P(4, 4) = 4! = 24$.
(e) ABC ends with C, and CDE starts with C, hence they must appear in sequence as $ABCDE$ as a unit, hence there are $P(3, 3) = 3! = 6$ permutations.
(f) 0, as BA (as in CBA) and BE (as in BED) can not happen simultaneously.

Ex. 6.3.16. Thirteen people on a softball team show up for a game.

(a) How many ways are there to choose 10 players to take the field?

(b) Of the 13 people who show up, three are women. How many ways are there to choose 10 players to take the field if at least one of these players must be a woman?

Solution:

(a) $C(13, 10) = 286$.

(b) $C(3, 1) \cdot C(10, 9) + C(3, 2) \cdot C(10, 8) + C(3, 3) \cdot C(10, 7) = 285$.

Ex. 6.3.19. How many 4-permutations of the positive integers not exceeding 100 contain three consecutive integers $k, k + 1, k + 2$, in the correct order

(a) where they are in consecutive positions in the permutation?

(b) where these consecutive integer can perhaps be separated by other integers in the permutation?

Solution:

(a) k can be $1, 2, \ldots, 98$, i.e., 98 choices. For each k, one can choose any one out of the remaining 97 numbers, and place it either before or after the three consecutive numbers, hence overall $98 \cdot 97 \cdot 2 = 19012$. However, we double counted the cases of four consecutive numbers, and there are 97 of them, hence the solution is $19012 - 97 = 18915$.

(b) Similarly, there are 98 choices of k, and one can choose any one out of the remaining 97 numbers, and place it in 4 possible positions, hence overall $98 \cdot 97 \cdot 4 = 38024$. Again we double counted the cases of four consecutive numbers, hence the solution is $38024 - 97 = 37927$.

Ex. 6.3.25. How many bit strings contain exactly eight 0's and 10 1's if every 0 must be immediately followed by a 1?

Solution: Since every 0 must be immediately followed by a 1 and there are eight 0's, so there are eight 01's. So there are 9 places to replace the remaining two 1's (7 places in-between + 2 places before the first and after the last 01). If the two 1's are placed together, there are $C(9, 1) = 9$ choices, if not, there are $C(9, 2)$=36 choices, hence together $9 + 36 = 45$ choices.

The questions below are optional. Attempt them to challenge yourself, but we will not be discussing them.

Additional Question. [Hard, IMO 2001] Suppose that 21 girls and 21 boys enter a mathematics competition. Furthermore, suppose that each entrant solves at most 6 questions, and for every boy-girl pair, there is at least 1 question that they both solved. Show that there is a question that was solved by at least 3 girls and at least 3 boys.

Solution: We know there are at least $21^2 = 441$ connections between boy and girl. Let S be the set of such connections with $|S| \geq 441$. Now we define 3 subsets A, B, C, where in the subset A are the connections corresponding to the problems solved by at most 2 girls, in B are the connections corresponding to the problems solved by at most 2 boys, and in C are the connections corresponding to the problems solved by at least 3 girls and at least 3 boys. Note that each of the connection must be in at least one of three sets, and A and B need not be disjoint. Now, we are ready to prove $C \neq \emptyset$.
Assume $C = \emptyset$ and pick a random boy, as he has solved at most 6 problems and there are 21 connections between this boy and all girls, hence there are at most 5 (out of the 6) problems

solved by at most 2 girls (as it can NOT be all 6 problems, since by generalized pigeonhole principle there is at least one problem solved by at least $\lceil \frac{21}{6} \rceil = 4$ girls). That means that each boy contributes to A by at most $5 \cdot 2 = 10$ connections. Therefore we have $|A| \leq 21 \cdot 10 = 210$. Similarly, $|B| \leq 210$.

$$|S| = |A \cup B \cup C| = |A \cup B| = |A| + |B| - |A \cap B| \leq 210 + 210 = 420,$$

which contradicts with $|S| \geq 441$, hence $C \neq \emptyset$, i.e., there are some question solved by at least 3 girls and at least 3 boys.

Additional Questions on Poker Cards. Five card poker is good setting in which we can practice our counting skills. In case you have never been to a casino, a card deck consists of 52 cards; each card has one of the 13 face values 2 3 4 5 6 7 8 9 10 J Q K A and one of the four suits $\clubsuit, \diamondsuit, \heartsuit, \spadesuit$. In five card poker, you are dealt a hand consisting of five different cards, for example

$$2\spadesuit, 9\heartsuit, K\diamondsuit, A\heartsuit, 2\heartsuit$$

and you win a prize if your hand is of a special type. We'll apply counting rules to figure out the number of different types of hands.

Four-of-a-kind. A hand is a *four-of-a-kind* if it contains four cards with the same value, for example:

$$\{J\heartsuit, J\diamondsuit, J\clubsuit, J\spadesuit, 2\diamondsuit\}.$$

Full house. A hand is a *full house* if it consists of three cards with one face value and two cards with another face value, for example,

$$\{8\spadesuit, 8\diamondsuit, 8\clubsuit, Q\diamondsuit, Q\spadesuit\}.$$

Flush. A hand is a *flush* if all five cards are of the same suit,[1] for example,

$$\{4\clubsuit, A\clubsuit, 9\clubsuit, K\clubsuit, 7\clubsuit\}.$$

Two-pair. A hand is a *two-pair* if it has two cards of one face value, two cards of another face value, and a fifth card of yet another face value, for example,

$$\{A\spadesuit, A\clubsuit, 8\diamondsuit, 8\clubsuit, 9\heartsuit\}.$$

Note that the order of the cards does not matter in a hand.

(a) How many *four-of-a-kind* hands are there?
(b) How many *full house* hands are there?
(c) How many *flush* hands are there?
(d) How many *two-pair* hands are there?

Solution:

(a) We can specify a four-of-a-kind sequence completely and uniquely by giving the face value of the four-of-a-kind, the face value of the fifth card, and the suit of the fifth card. There are 13 choices for the face value of the four-of-a-kind. Each of them leaves out 12 choices for the face value of the fifth card and 4 choices for its suit. By the generalized product rule, the number of four-of-a-kind hands is

$$13 \cdot 12 \cdot 4 = 624.$$

[1] In poker there is also a "royal flush" and a "straight flush" that are sometimes counted separately. We will include those into our count of flushes.

(b) We can specify each full house completely and uniquely by giving the face value of the cards in the triple, the suits of the cards in the triple, the face value of the cards in the pair, and the suits of the cards in the pair. There are 13 choices for the face value of the triple, $C(4,3)$ choices for the suits in the triple (three suits out of a set of four), 12 remaining choices for the face value of the pair, and $C(4,2)$ choices for the suits of the cards in the pair. By the generalized product rule, the number of full house hands is

$$13 \cdot C(4,3) \cdot 12 \cdot C(4,2) = 13 \cdot 4 \cdot 12 \cdot 6 = 3,744.$$

(c) We can specify a flush uniquely by describing the suit of all the cards in it and their face values. The suit can be chosen in 4 ways and the five face values can be chosen in $C(13,5)$ ways. By the product rule, the number of flushes is

$$C(13,5) \cdot 4 = 5,148.$$

(d) There are $C(13,2)$ choices for the face values of the two pairs. Once these face values have been chosen, there are $C(4,2)$ choices for the suits of the smaller pair, and again $C(4,2)$ choices for the suits of the bigger pair. Finally, this leaves out 11 choices for the face value of the last card and 4 choices for its suit, giving a total of

$$C(13,2) \cdot C(4,2) \cdot C(4,2) \cdot 11 \cdot 4 = 123,552$$

possible two-pair hands.

Additional Question. Use the inclusion-exclusion principle to answer the following question: How many permutations of the 10 letters $\{a,b,c,d,e,f,g,h,i,j\}$ are there that do not contain any of the patterns hi, fad, and jig? For example, the permutations $hedafjigcb$ and $jghiebfadc$ should not be counted because the first one contains jig and the second one contains both hi and fad.

Solution: Let A, B and C be the sets of permutations that contain a hi, a fad, and a jig, respectively. We will first calculate the size of $A \cup B \cup C$ using the inclusion-exclusion principle. How many elements does A have? This set contains the permutations in which hi must appear in sequence so we can think of these two letters as a single "symbol" that is permuted with the 8 other letters. This way we can view A as the set of permutations of the 9-element set $\{a,b,c,d,e,f,g,hi,j\}$ and $|A| = 9!$. Similarly we get $|B| = 8!$ and $|C| = 8!$. The set $A \cap B$ contains those permutations that contain both $hiandfad$. We can view $A \cap B$ as the set of permutations of $\{b,c,e,g,j,hi,fad\}$, so $|A \cap B| = 7!$. Similarly, $|B \cap C| = 6!$. The set $A \cap C$ is empty because no permutation contains both hi and jig — the i must appear exactly once. Plugging into the inclusion-exclusion principle, we get $|A \cup B \cup C| = 9! + 8! + 8! - 7! - 6! = 437,760$.
We want to know how many permutations *do not* contain a hi, a fad, or a jig. This is the complement of the set $A \cup B \cup C$, so the desired number is

$$|\overline{A \cup B \cup C}| = 10! - |A \cup B \cup C| = 3,628,800 - 437,760 = 3,191,040.$$